

Scientific Computation

Federico Formica
Material by: Prof. Matthew Giamou
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Lecture 25 - Interpolation Errors

Lecture Outline

- Newton's interpolating example

- Polynomial interpolation error
- Chebyshev nodes

Divided Differences

Given $x_0, x_1, \dots, x_n$ where $0 \leq i < j \leq n$, recursively define

$$[y_i, \dots, y_j] = \frac{[y_{i+1}, \dots, y_j] - [y_i, \dots, y_{j-1}]}{x_j - x_i},$$

$$[y_i] = y_i,$$

where $[y_i, \dots, y_j]$ are called **divided differences** over $x_i, \dots, x_j$.

↳ this notation simplifies our Newton form's coefficient

$$\text{E.g. } [y_1, y_2, y_3] = \frac{[y_2, y_3] - [y_1, y_2]}{x_3 - x_1}$$

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Newton Coefficients from Divided Differences

Using this definition, we can write

$$c_0 = y_0 = [y_0]$$

$x_1 - x_0$

$$c_1 = \frac{y_1 - y_0}{x_1 - x_0} = [y_0, y_1]$$

$$c_2 = \frac{\frac{y_2 - y_1}{x_2 - x_1} - \frac{y_1 - y_0}{x_1 - x_0}}{x_2 - x_0} = \frac{[y_1, y_2] - [y_0, y_1]}{x_2 - x_0} = [y_0, y_1, y_2]$$

⋮

$$c_n = \frac{[y_1, \dots, y_n] - [y_0, \dots, y_{n-1}]}{x_n - x_0} = [y_0, y_1, \dots, y_n]$$

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Computing Newton Coefficients

Example 1: Determine the Newton polynomial for the dataset

$$x_i : \quad 1 \quad 2 \quad 4$$

$$y_i : \quad 1 \quad 3 \quad 3$$

i	x_i	$[y_i] = y_i$	$[y_{i-1}, y_i]$	$[y_{i-2}, y_{i-1}, y_i]$
0	1	1	N/A	N/A
1	2	3	$[y_0, y_1]$	N/A
2	4	3	$[y_1, y_2]$	$[y_0, y_1, y_2]$

↳ Our Newton polynomial's coeffs are on the diagonal

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Computing Newton Coefficients

Example 1 cont.: Let's compute the coefficients by hand:

i	x_i	$[y_i] = y_i$	$[y_{i-1}, y_i]$	$[y_{i-2}, y_{i-1}, y_i]$
0	1	1	N/A	N/A
1	2	3	$\underline{[y_0, y_1] = 2}$	N/A
2	4	3	$\underline{[y_1, y_2] = 0}$	$[y_0, y_1, y_2] = -\frac{2}{3}$

$$[y_0, y_1] = \frac{y_1 - y_0}{x_1 - x_0} = \frac{3-1}{2-1} = 2$$

$$[y_0, y_1, y_2] = \frac{[y_1, y_2] - [y_0, y_1]}{x_2 - x_0}$$

$$[y_1, y_2] = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3-3}{4-2} = 0$$

$$= \frac{0-2}{4-1} = -\frac{2}{3}$$

$$p_2(x) = [y_0] + [y_0, y_1](x - x_0) + [y_0, y_1, y_2](x - x_0)(x - x_1)$$

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Computing Newton Coefficients

Example 1 cont.: Suppose we add in a fourth point:

$$x_i: \quad 1 \quad 2 \quad 4 \quad 3$$

$$y_i: \quad 1 \quad 3 \quad 3 \quad 5$$

yes!

Can we leverage our pre-computed results for the first three points? Let's look at the definition of our Newton polynomial basis:

$$p_n(x) = c_0 + c_1 \underbrace{(x - x_0)}_{\phi_0=1} + c_2 \underbrace{(x - x_0)(x - x_1)}_{\phi_2} + \dots c_n \underbrace{(x - x_0)(x - x_1) \dots (x - x_{n-1})}_{\phi_n}$$

And a reminder that our coefficients are:

$$+ c_{n+1} (x - x_0) \dots (x - x_{n-1})$$

$$c_n = \frac{[y_1, \dots, y_n] - [y_0, \dots, y_{n-1}]}{x_n - x_0} = [y_0, y_1, \dots, y_n]$$

$\Rightarrow \phi_n$ and c_n don't depend on data

x_k, y_k for $k > n$.

$\hookrightarrow$ does not depend on points that arrive after n .

the prior basis functions don't depend on x_n .

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Computing Newton Coefficients

Example 1 cont.: The old coefficients remain the same! *Easy to update*

$$x_i: 1 \quad 2 \quad 4 \quad 3$$

$$y_i: 1 \quad 3 \quad 3 \quad 5$$

$O(n)$

i	x_i	$[y_i] = y_i$	$[y_{i-1}, y_i]$	$[y_{i-2}, y_{i-1}, y_i]$	$[y_{i-3}, y_{i-2}, y_{i-1}, y_i]$
0	1	1	N/A	N/A	N/A
1	2	3	2	N/A	N/A
2	4	3	0	$-\frac{2}{3}$	N/A
3	3	5	-2	-2	$-\frac{2}{3}$

$$[y_2, y_3] = \frac{[y_3] - [y_2]}{x_3 - x_2} = \frac{5 - 3}{3 - 4} = -2$$

$$[y_1, y_2, y_3] = \frac{-2 - 0}{3 - 2} = -2$$

$$[y_0, y_1, y_2, y_3] = \frac{-2 - (-\frac{2}{3})}{3 - 1} = -\frac{2}{3}$$

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Polynomial Interpolation's Assumptions

Recall the general setting we are working with:

- We have $n + 1$ **distinct** points $\{(x_i, y_i)\}_{i=0}^n$ in our dataset
- There is a unique interpolating polynomial p_n of degree at most n .
- We are enforcing $f(x_i) = p_n(x_i) = y_i$

Let's add in one additional assumption:

$$f^{(n+1)}(x) \text{ is continuous on } [a, b] \supset \{x_i\}_{i=0}^n$$

$$\Leftrightarrow C^{n+1}[a, b]$$

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A Polynomial Error Bound

With this new assumption, we can prove the following theorem:

Theorem 1: If $p_n(x)$ is the unique polynomial that interpolates f for $n + 1$ distinct nodes x_i and $f^{(n+1)}$ is continuous, then for any $x \in [a, b]$, there exists $\xi = \xi(x) \in (a, b)$ such that

$$f(x) - p_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^n (x - x_i).$$

$\xi = \xi(x)$ is satisfied by assumption

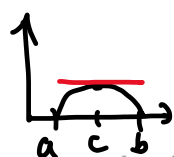
Proof: Assume $\lambda \neq \alpha$, $\forall i = 0, \dots, n$. (This is not the case if $\lambda = \alpha$)

For this λ , define $w(t) = \prod_{i=0}^n (t - x_i)$ polynomial in t

and $C \triangleq \frac{f(\lambda) - p_n(\lambda)}{w(\lambda)}$ (C is constant w.r.t. t)

Finally, let $\varphi(t) \triangleq f(t) - p_n(t) - Cw(t)$
 $\hookrightarrow$ depends on λ .

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Proof of Theorem 1 Cont.

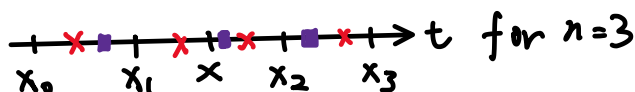
The function $\varphi(x)$ takes the value of 0 at $n+2$ points: $x_0, x_1, \dots, x_n$ and α .

Next, we invoke Rolle's Theorem:

$\hookrightarrow$ Crucial that $x_i \neq x_j$

Theorem 2: For a function f that is continuous on $[a, b]$ and differentiable on (a, b) , if $f(a) = f(b) = 0$, then $f'(c) = 0$ for some $c \in (a, b)$.

Rolle's thm: $\varphi'(t) = 0$ for some t between each pair of the roots of φ :



$\Rightarrow \varphi'$ has $n+1$ distinct roots!

$= \varphi'$ has n distinct roots

$\vdots$
 $\varphi^{(n+1)}$ has at least one root

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Proof of Theorem 1 Cont.

Let ξ be a root of $\varphi^{(n+1)}$. Our repeated application of Rolle's theorem ensures $\xi \in (a, b)$. We have:

$$0 = \varphi^{(n+1)}(\xi) = f^{(n+1)}(\xi) - p^{(n+1)}(\xi) - Cw^{(n+1)}(\xi).$$

Since the degree of p_n is $\leq n \Rightarrow p_n^{(n+1)}(t) = 0 \quad \forall t$

$w(t) = \prod_{i=0}^n (t-x_i)$ has only one term of degree $n+1$:

$$\frac{d^{n+1}}{dt^{n+1}} (t^{n+1}) = (n+1)! \quad (\text{repeat differentiation})$$

$$\text{Thus, } f^{(n+1)}(\xi) - C(n+1)! = f^{(n+1)}(\xi) - \frac{f(x) - p_n(x)}{w(x)} (n+1)!$$

$$\Rightarrow f(x) - p_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^n (x - x_i)$$

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Proof of Theorem 1 Cont.

Going back to the original equation, we can rewrite it as:

$$0 = \varphi^{(n+1)}(\xi) = f^{(n+1)}(\xi) - p^{(n+1)}(\xi) - c w^{(n+1)}(\xi).$$

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Applying Theorem 1: Even Spacing

If $M = \max_{t \in [a, b]} |f^{(n+1)}(t)|$, then we have

$$a + nh = b$$

$$|f(x) - p_n(x)| \leq \frac{M}{(n+1)!} \prod_{i=0}^n |x - x_i|.$$

Suppose we **evenly space** the points: $x_i = a + ih$ for $i = 0, 1, \dots, n$, where $h = (b - a)/n$. Then we can use the following lemma:

$$\prod_{i=0}^n |x - x_i| \leq \frac{1}{4} h^{n+1} n!$$

$$\Rightarrow |f(x) - p_n(x)| \leq \frac{M}{4(n+1)!} h^{n+1} (n!) \\ = \frac{M}{4(n+1)} h^{n+1}$$

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Divided difference的意思就是差商，它可以看作是离散版本对导数的一个近似，一般来说，k order divided difference 可以被表示为 $\frac{f^k(x)}{k!}$

所以你会发现，Newton interpolation实际上就是算差分，且其加新点不会影响前面的计算。

1. 第一部分：多项式插值的基本设定

(这三条是所有插值基本前提)

✓ (1) 有 $n + 1$ 个互不相同的点

$$\{(x_i, y_i)\}_{i=0}^n, \quad x_i \neq x_j \text{ for } i \neq j$$

为什么必须 distinct?

- 如果两个点的 x 一样但 y 不一样 $\rightarrow$ 没有多项式能同时满足
- 如果 x 一样 y 一样 $\rightarrow$ 点重复，不影响插值，但会造成算法退化

所以 所有 x_i 必须不同 —— 这是唯一性和可解性的基础。

✓ (2) 存在且唯一一个 n 次或以下的多项式 $p_n(x)$

这是经典定理：

对于 $n+1$ 个不同的 x 点，存在且唯一一个次数 $\leq n$ 的多项式通过所有点。

也就是说：

- Lagrange 插值公式能构造出这个多项式
- Newton 插值公式也能构造出
- 实质上这两个公式构造出的都是同一个多项式

唯一性来自线性代数中 Vandermonde 矩阵可逆性。

✓ (3) 我们要求插值满足：

$$p_n(x_i) = f(x_i) = y_i$$

就是说：

- 你提供的 y 数据点，就是插值多项式的取值
- 插值是 exact fitting（精确通过所有点）

■ 2. 为什么加一个额外假设： $f^{(n+1)}(x)$ 连续？

这一行：

$$f^{(n+1)}(x) \text{ is continuous on } [a, b]$$

等价于：

$$f \in C^{n+1}[a, b]$$

意思是：

f 有 $n+1$ 阶连续导数。

■ 3. 这条假设（ f 的 $n+1$ 阶导数连续）是为了什么？

不是为了插值多项式的存在性或唯一性。

它是为了插值误差估计。

你之后一定会看到插值误差公式：

$$f(x) - p_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)(x - x_1) \cdots (x - x_n)$$

这个公式要成立，就必须要求：

- $f^{(n+1)}(\xi)$ 存在
- 而且是连续的

否则：

- 你不能保证误差公式里的 $f^{(n+1)}(\xi)$ 有意义
- 能不能用拉格朗日形式表达这个余项都是问题

► 更直观理解

误差项来自泰勒展开或者多变量版的中值定理：

$$f[x_0, \dots, x_n, x] = \frac{f^{(n+1)}(\xi)}{(n+1)!}$$

这个等式必须用到 f 在区间里具有 连贯的 $n+1$ 阶变化信息（即 $n+1$ 阶导数连续）。

■ 4. 图中的符号解释： $C^{n+1}[a, b]$

这是数学中标准的函数空间表示：

- C^0 ：连续函数
- C^1 ：一阶导数连续
- C^2 ：二阶导数连续
- ...
- C^{n+1} ： $n+1$ 阶导数连续

所以：

$$f^{(n+1)}(x) \text{ continuous on } [a, b] \iff f \in C^{n+1}[a, b]$$

两者是完全等价的写法。

5. 总结（最重要）

图里总结的是多项式插值的全部假设：

必需假设（保证插值存在唯一）：

1. 有 $n + 1$ 个不同的点 x_i
2. 插值多项式要满足 $p_n(x_i) = y_i$

附加假设（不是为了插值本身，而是为了误差公式）：

3. $f \in C^{n+1}[a, b]$
 - f 的 $n+1$ 阶导数存在且连续
 - 插值误差可以用 $f^{(n+1)}(\xi)$ 来表达

后面的定理给出了存在一个点 ξ ，让插值函数在非插值点的误差等于一个公式。

均匀取点会导致误差爆炸，最好不要均匀取点。

如果节点均匀分布，那么插值误差项的乘积项

$$\prod |x - x_i|$$

可以大到 $\sim h^{n+1} n!$,

导致误差急剧增大，使插值发散。

这就是 Runge 现象的数学本质。